

Variability in Pesticide Deposition and Source Contributions to Snowpack in Western US National Parks

Fifty-six seasonal snowpack samples were collected at remote alpine, sub-arctic, and arctic sites in eight Western US national parks during three consecutive years (2003–2005). Four current-use pesticides (CUPs) (dacthal (DCPA), chlorpyrifos, endosulfan, and γ -hexachlorocyclohexane (HCH)) and four historic-use pesticides (HUPs) (dieldrin, α -HCH, chlordane, and hexachlorobenzene (HCB)) were commonly measured at all sites, during all years. The mean coefficient of variation for pesticide concentrations was 15% for site replicate samples, 41% for intra-park replicate samples, and 59% for inter-annual replicate samples. The relative pesticide concentration profiles were consistent from year to year but unique for individual parks, indicating a regional source effect. HUP concentrations were well-correlated with regional cropland intensity when the effect of temperature on snow-air partitioning was considered. The mass of individual CUPs used in regions located one-day upwind of the parks was calculated using air mass back trajectories and this was used to explain the distribution of CUPs among the parks. The percent of the snowpack pesticide concentration due to regional transport was high (>75%) for the majority of pesticides in all parks. These results suggest that the majority of pesticide contamination in US national parks is due to pesticide use in North America.